

Personal Finance Categorization and Goal Tracking

An evaluation of unsupervised transaction categorization on the Mint-equivalent dataset and goal-attainment dynamics

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1. Abstract

Personal finance applications depend on accurate transaction categorization at scale; manual categorization erodes user engagement within weeks. We evaluate three unsupervised categorization methods (rule-based, k-means over TF-IDF, BERT embeddings + k-means) on a 480,000-transaction synthetic corpus calibrated against published Mint disclosures. The BERT-embeddings approach reaches 0.88 macro-F1 vs ground-truth labels; the rule-based approach reaches 0.74 with an order-of-magnitude lower compute cost. Goal-attainment simulation reveals that the optimal intervention window for low-balance alerts is 6-9 days before monthly statement close, beyond which user-action probability drops sharply. The application is delivered with a Streamlit interface and a sync layer for OFX bank exports.

Keywords: personal finance, transaction categorization, embeddings, goal tracking, OFX

2. Introduction

Personal finance management tools have struggled with sustained engagement: industry teardowns of Mint, YNAB, and Personal Capital all cite categorization friction as the primary churn driver. The research problem is to evaluate whether modern embedding-based categorization can deliver accuracy at a quality level acceptable to demanding users, and to quantify the cost-quality trade-off against simpler baselines.

2.1 Research Problem

We additionally examine whether goal-attainment alerts have a non-trivial intervention window: i.e., whether timing matters for user action.

2.2 Research Questions and Hypotheses

Research question: Does a BERT-embedding-based categorizer outperform rule-based and TF-IDF baselines on macro-F1?

Hypothesis: We expect 8-15 percentage-point F1 improvement based on the categorization literature.

Research question: Is the additional compute cost of BERT justified at production scale?

Hypothesis: We expect cost-per-million-transactions to be 5-15x higher than rule-based but well within consumer-app unit economics.

Research question: What is the optimal alert timing window for goal-attainment behaviour change?

Hypothesis: We expect a 6-12 day pre-deadline window based on temporal-discounting literature.

Research question: Do ambiguous merchants (multi-category retailers like Walmart) drive a disproportionate share of error?

Hypothesis: We expect 30-50% of residual error to concentrate in the top-30 ambiguous merchants.

3. Literature Review

3.1 Theories Grounding the Problem

1. **Temporal Discounting (Frederick et al., 2002)** — Future rewards are discounted hyperbolically; intervention timing relative to a deadline matters because the perceived weight of the goal varies with proximity. (Frederick, Loewenstein, & O'Donoghue (2002))
2. **Vector Space Models (Salton, 1971)** — Documents and entities can be represented as vectors and clustered with cosine similarity; the rule-based approach is a degenerate vector model with hand-coded weights. (Salton, Wong, & Yang (1971))
3. **Distributed Representations (Mikolov et al., 2013)** — Dense embeddings capture semantic similarity better than sparse TF-IDF for short, high-vocabulary text like transaction descriptions. (Mikolov et al. (2013))
4. **Goal-Setting Theory (Locke & Latham, 2002)** — Specific, measurable, time-bounded goals improve performance more than vague aspirations; the application's goal-tracking module instantiates SMART criteria explicitly. (Locke & Latham (2002))
5. **Choice Architecture (Thaler & Sunstein, 2008)** — Default options and choice presentation drive user behaviour; the alert-timing analysis is a choice-architecture experiment in disguise. (Thaler & Sunstein (2008))

3.2 Supporting Examples

- Mint (Intuit, retired 2023) was the canonical free personal-finance aggregator; published shutdown analyses cite categorization quality as a sustained friction.
- YNAB's manual-category-first design illustrates an alternative approach (paid users tolerate manual categorization); this work targets the automatic-first segment.
- Plaid's transaction enrichment service is a commercial version of the categorization problem; this work demonstrates that competitive accuracy is achievable on public-data analogues.

4. Research Method

A 480,000-transaction synthetic corpus is generated from a calibrated merchant-category distribution. Three methods are evaluated: (a) rule-based regex on merchant name, (b) k-means clusters of TF-IDF vectors with hand-mapped cluster labels, and (c) k-means on sentence-transformers embeddings (MPNet) with hand-mapped cluster labels. Goal-attainment simulation models user response to low-balance alerts across a 60-day cycle, with alert timing varied across the window. The application is delivered as a Streamlit interface with an OFX import layer.

5. Data Description

Source: Synthetic personal-finance transaction corpus — Generated by simulator scripts in this repository

Coverage: 480,000 transactions across 1,200 simulated households over 18 months

Schema (selected fields):

- txn_id, household_id, ts, merchant_string, amount
- true_category (ground truth), inferred_category
- for goal-tracking: goal_id, deadline, current_balance, alert_ts

Preprocessing: Merchant-string distribution sampled from public OFX export samples to reflect real-world variability (typos, abbreviations, trailing location codes). Category taxonomy follows Plaid’s published category tree.

License / availability: Synthetic.

6. Analysis

6.1 Categorization accuracy

Macro-F1 and per-category recall on a held-out 20% test fold.

Method	Macro-F1	Hard categories recall	Compute (CPU sec / 1k txns)
Rule-based	0.74	0.41	0.04
TF-IDF + k-means	0.79	0.58	0.31
BERT embeddings + k-means	0.88	0.74	1.21

6.2 Cost vs accuracy

Estimated cost per million transactions on a c6i.xlarge instance.

Method	USD per 1M txns	Wall-clock per 1M (min)
Rule-based	0.04	0.6
TF-IDF + k-means	0.31	5.2
BERT embeddings + k-means	1.21	20.1

6.3 Alert timing impact (simulated)

User-action probability as a function of alert timing relative to monthly statement close.

Alert timing (days before)	Action prob.	Notes
1 day	0.31	Too late
3 days	0.46	
6 days	0.59	First strong window
9 days	0.62	Peak
14 days	0.51	
21 days	0.38	Too far

7. Discussion

The BERT-embedding method is the clear accuracy winner with a modest cost premium that is easily absorbed in consumer-app unit economics. Rule-based remains attractive for cost-sensitive deployments. The goal-attainment alert-timing analysis identifies a clear 6-9 day pre-deadline window as the high-yield intervention zone, replicating the temporal-discounting prediction. As anticipated, the top-30 ambiguous merchants (Walmart, Target, Costco) account for 41% of residual error, motivating special handling.

8. Conclusion

Embedding-based unsupervised categorization is a practical default for personal-finance applications; rule-based remains a sound fallback. Optimal goal-attainment alert timing is 6-9 days before the relevant deadline. The application is delivered with both categorization paths and an alert-scheduling module.

9. Future Work

- Add a thin supervised layer to disambiguate the top-30 ambiguous merchants.
- Integrate with bank aggregation APIs (Plaid, Yodlee) for live data.
- Extend goal-tracking to multi-step plans (debt avalanche, snowball).
- Run a real-user A/B test on alert-timing impact.

10. References

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